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FAUCET DEVICE

Abstract

A faucet device includes a water discharge part that discharges a hot or cold water and a faucet main body that is arranged inside a bathroom and supplies a hot or cold water to the water discharge part. The faucet main body includes a flow-rate/temperature control part that executes flow rate control and temperature control for a hot or cold water that is discharged from the water discharge part, a motor that drives the flow-rate/temperature control part, a controller that controls driving of the motor, a first waterproof case that houses the motor and isolates the motor from an internal space of the bathroom, and a second waterproof case that houses the controller and isolates the controller from an internal space of the bathroom. The first waterproof case and the second waterproof case are communicated with an external space of the bathroom.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is based upon and claims the benefit of priority to Japanese Patent Application No. 2024-036270, filed on Mar. 8, 2024, the entire contents of which are herein incorporated by reference.

FIELD
[0002] A disclosed embodiment(s) relate(s) to a faucet device.
BACKGROUND
[0003] A faucet device has conventionally been known where a motor that drives a temperature control part and a flow rate control part for a hot or cold water and a controller that controls driving of such a motor are housed in a waterproof case so as to be isolated from an external space of a bathroom (see, for example, Japanese Patent Application Publication No. H03-275822).
[0004] However, in a conventional faucet device as described above, dew condensation may occur inside a waterproof case that houses a motor and a controller. Furthermore, in a conventional faucet device as described above, a motor and a controller are housed in a single waterproof case, so that a size of such a waterproof case may be increased needlessly, so as to cause a size increase of a faucet main body.
SUMMARY
[0005] According to an aspect of an embodiment, a faucet device includes a water discharge part that discharges a hot or cold water, and a faucet main body that is arranged inside a bathroom and supplies a hot or cold water to the water discharge part, wherein the faucet main body includes a flow-rate/temperature control part that executes flow rate control and temperature control for a hot or cold water that is discharged from the water discharge part, a motor that drives the flow-rate/temperature control part, a controller that controls driving of the motor, a first waterproof case that houses the motor and isolates the motor from an internal space of the bathroom, and a second waterproof case that houses the controller and isolates the controller from an internal space of the bathroom, and the first waterproof case and the second waterproof case are communicated with an external space of the bathroom.

Description

BRIEF DESCRIPTION OF DRAWING(S)
[0006] FIG. **1** is a diagram that illustrates an example of a bathroom unit where a faucet device according to an embodiment is arranged therein.
[0007] FIG. **2** is a front view that illustrates a remote controller.
[0008] FIG. **3** is a block diagram that illustrates an outline of a faucet device according to an embodiment.
[0009] FIG. **4** is a perspective view (part 1) that illustrates a faucet main body.
[0010] FIG. **5** is a perspective view that illustrates a water mixing faucet unit.
[0011] FIG. **6** is a front view that illustrates a water mixing faucet unit.
[0012] FIG. **7** is a perspective view thereof in a cross section along VII-VII in FIG. **6**.
[0013] FIG. **8** is a perspective view (part 2) of a faucet main body.
[0014] FIG. **9** is a perspective view that illustrates a first waterproof case and a second waterproof case.

[0015] FIG. **10** is a perspective view that illustrates a variation of a first waterproof case and a second waterproof case.
[0016] FIG. **11** is a cross-sectional view that illustrates a binding member and a sealing member.
[0017] FIG. **12** is a perspective view that illustrates a binding member and a sealing member.
[0018] FIG. **13** is a perspective view that illustrates a cover member.

DESCRIPTION OF EMBODIMENT(S)

[0019] Hereinafter, an embodiment(s) of a faucet device as disclosed in the present application will be explained in detail, with reference to the accompanying drawing(s). Additionally, this invention is not limited by an embodiment(s) as illustrated below.
[0020] A faucet device **1** according to an embodiment is provided in, for example, a bathroom unit **1000** that is a bathroom, as illustrated in FIG. **1**. FIG. **1** is a diagram that illustrates an example of a bathroom unit **1000** where a faucet device **1** according to an embodiment is arranged therein.
[0021] The bathroom unit **1000** includes a bathtub **1001**, a first counter **1002**, a second counter **1003**, and the faucet device **1**. Additionally, in a case where a cold water that is discharged by the faucet device **1** and a mixed water that is provided by mixing a hot water and a cold water and is discharged by the faucet device **1** are not distinguished, they will be referred to and explained as a “hot or cold water” below.
[0022] Hereinafter, an explanation may be provided by using frontward and backward directions where a direction of projecting of the second counter **1003** from a wall part **1004***a* of the bathroom unit **1000** is a frontward direction and an opposite side of such a frontward direction is a backward direction, and further, upward and downward directions where a vertical direction is a downward direction and an opposite side of such a downward direction is an upward direction. Furthermore, an explanation may be provided where directions that are orthogonal to frontward and backward directions and upward and downward directions are leftward and rightward directions.
[0023] The first counter **1002** is attached to the wall part **1004***a* of the bathroom unit **1000**. The first counter **1002** projects from the wall part **1004***a* to an inside of a bathroom. The first counter **1002** is provided on an upper side of a washing place floor **1005** of the bathroom unit **1000**. The first counter **1002** houses a faucet main body **3** of the faucet device **1**.
[0024] The second counter **1003** is attached to the wall part **1004***a*. The second counter **1003** projects from the wall part **1004***a* to an inside of a bathroom. The second counter **1003** is provided on an upper side of the first counter **1002**. For example, the second counter **1003** is provided so as to extend to an upper side of the bathtub **1001**. Additionally, the second counter **1003** does not have to extend to an upper side of the bathtub **1001**.
[0025] The second counter **1003** houses a part of the faucet device **1**. Specifically, the second counter **1003** houses a part of a water tap **10** of the faucet device **1** and a part of a hand shower **11** of the faucet device **1**. The second counter **1003** is provided so as to expose a water discharge port **10***a* of the water tap **10**. The water discharge port **10***a* of the water tap **10** is provided on the second counter **1003** in such a manner that a direction of discharging of a hot or cold water is a downward direction.
[0026] Furthermore, the second counter **1003** is connected to a shower hose **11***a* of the hand shower **11** of the faucet device **1**. The shower hose **11***a* is connected to a shower water conduit that is housed in the second counter **1003**.
[0027] An overhead shower **12** of the faucet device **1** and a warm pillar **13** of the faucet device **1** are attached to a ceiling **1006** of the bathroom unit **1000**. The overhead shower **12** and the warm pillar **13** are provided integrally.
[0028] The overhead shower **12** discharges a hot or cold water to an area of a user that is greater than that for the hand shower **11**. For example, the overhead shower **12** is configured to apply a hot or cold water to a whole body of a user. The warm pillar **13** integrates a hot or cold water into, directs a flow of, and discharges a single stream of water. In other words, the warm pillar **13** directs a flow of, and discharges, a hot or cold water in such a manner that an uninterrupted hot or cold

water flows down like a pillar.

[0029] A remote controller **4** (an operation part) of the faucet device **1** is attached to a wall part **1004***b* of the bathroom unit **1000**. The remote controller **4** may be attached to the wall part **1004***a* where the first counter **1002** and the second counter **1003** are attached thereto.

[0030] The remote controller **4** receives various types of operations of a user for the faucet main body **3**. Specifically, the remote controller **4** receives a setting operation for a temperature of a hot or cold water on the faucet main body **3**. The remote controller **4** receives a setting operation for a flow rate of a hot or cold water on the faucet main body **3**. The remote controller **4** receives a switching operation for discharging and stopping of a hot or cold water. The remote controller **4** receives a switching operation for a discharging place of a hot or cold water. In a case where the remote controller **4** is operated by a user, it transmits an operation signal that corresponds to each operation to a controller **7** (see FIG. **3**) of the faucet device **1**.

[0031] The remote controller **4** includes, for example, a temperature control button **41**, a water flow rate control button **42**, and a switching button **43**, as illustrated in FIG. **2**. FIG. **2** is a front view that illustrates a remote controller **4**.

[0032] The temperature control button **41** is a button for controlling a temperature of a mixed water in a faucet main body **3**. The temperature control button **41** includes a high temperature button **41***a* and a low temperature button **41***b*. The high temperature button **41***a* is a button for raising a temperature of a mixed water. The low temperature button **41***b* is a button for lowering a temperature of a mixed water. Additionally, the remote controller **4** displays a setting state for a temperature of a mixed water on a first display part **45***a*. In a case where the high temperature button **41***a* or the low temperature button **41***b* is operated, a display of the first display part **45***a* is changed depending on an operation of each button **41***a*, **41***b*.

[0033] Additionally, a temperature of a mixes water in the faucet main body **3** is controllable within a predetermined temperature range. In a case where the low temperature button **41***b* is operated in such a manner that a set temperature of a mixed water is lower than a minimum temperature within a predetermined temperature range, a hot water is not mixed into a cold water in the faucet main body **3**, so that such a cold water is discharged.

[0034] The water flow rate control button **42** is a button for controlling a flow rate of a hot or cold water that is discharged from a faucet device **1**. The water flow rate control button **42** includes a water increasing button **42***a* and a water decreasing button **42***b*. The water increasing button **42***a* is a button for increasing a flow rate of a hot or cold water. The water decreasing button **42***b* is a button for decreasing a flow rate of a hot or cold water. Additionally, the remote controller **4** displays a setting state for a water flow rate of a hot or cold water on a second display part **45***b*. In a case where the water increasing button **42***a* or the water decreasing button **42***b* is operated, a display of the second display part **45***b* is changed depending on an operation of each button **42***a*, **42***b*. Additionally, a flow rate of a hot or cold water that is discharged from the faucet device **1** is controllable within a predetermined flow rate range.

[0035] The switching button **43** is a button for switching between discharging and stopping of a hot or cold water in the faucet device **1**. Furthermore, the switching button **43** is a button for switching a discharging place of a hot or cold water in the faucet device **1**. The switching button **43** includes a water tap button **43***a*, a hand shower button **43***b*, an overhead shower button **43***c*, and a warm pillar button **43***d*. Each of buttons **43***a* to **43***d* is pushed by a user, so as to be switched between "ON" and "OFF".

[0036] In a case where each of buttons **43***a* to **43***d* of the switching button **43** is "OFF", no hot or cold water is discharged. That is, the faucet device **1** is provided in a water stopping state thereof.

[0037] In a case where one switching button **43** is pushed from a state where each of buttons **43***a* to **43***d* of the switching button **43** is "OFF" and thereby a pushed switching button **43** is turned to "ON", a hot or cold water is discharged. That is, the faucet device **1** is turned from a water stopping state to a water discharging state thereof.

[0038] In a case where another switching button **43** is pushed in a state where one switching button **43** is "ON", a switching button **43** that is "ON" is changed, so that a discharging place of a hot or cold water is switched.
[0039] For example, in a case where the hand shower button **43***b* is "ON", a hot or cold water is discharged from a hand shower **11**. In a case where the water tap button **43***a* is pushed in such a state, the hand shower button **43***b* is turned to "OFF" and the water tap button **43***a* is turned to "ON". Thereby, a discharging place of a hot or cold water is changed from the hand shower **11** to a water tap **10**, so that such a hot or cold water is discharged from the water tap **10**.
[0040] In a case where a switching button **43** that is "ON" is pushed again, such a pushed switching button **43** is turned to "OFF", so that each of buttons **43***a* to **43***d* of the switching button **43** is "OFF" and a hot or cold water is stopped. That is, the faucet device **1** is turned from a water discharging state to a water stopping state thereof.
[0041] Additionally, each of the buttons **43***a* to **43***d* is configured in such a manner that it is possible for a user to recognize "ON" and "OFF" states thereof. For example, a switching button **43** that is "ON" is turned on and a switching button **43** that is "OFF" is turned off.
[0042] Additionally, although the faucet device **1** that is allowed to discharge a hot or cold water from an overhead shower **12**, a warm pillar **13**, the hand shower **11**, and the water tap **10** has been explained as an example herein, this is not limitative. For example, the faucet device **1** may be configured to have none of the overhead shower **12** and the warm pillar **13**.
[0043] Next, an outline of a faucet device **1** according to an embodiment will be explained with reference to FIG. **3**. FIG. **3** is a block diagram that illustrates an outline of a faucet device **1** according to an embodiment. In FIG. **3**, a flow of a hot or cold water is indicated by an arrow of a solid line and a communication line is indicated by a broken line.
[0044] The faucet device **1** includes a plurality of water discharge parts **2**, a faucet main body **3**, a remote controller **4**, and a communication unit **5**.
[0045] The plurality of water discharge parts **2** include a water tap **10**, a hand shower **11**, an overhead shower **12**, and a warm pillar **13**.
[0046] The faucet main body **3** includes a water mixing faucet unit **20**, a water discharge switching part **30**, and a controller **7**.
[0047] The water mixing faucet unit **20** includes a flow-rate/temperature control part(s) that execute(s) flow rate control and temperature control for a hot or cold water that is discharged from a water discharge part(s) **2**. The water mixing faucet unit **20** includes a hot and cold water mixing part **50** that is a temperature control part for a hot or cold water, and a flow rate control part **60**, as flow-rate/temperature control parts. A hot water is supplied from a hot water supply source **80** to the hot and cold water mixing part **50**. Furthermore, a cold water is supplied from a cold water supply source **81** to the hot and cold water mixing part **50**. A water shut-off valve **84** is provided in a flow channel (a hot water supply pipe **82**) between the hot and cold water mixing part **50** and the hot water supply source **80**. Furthermore, a water shut-off valve **85** is provided in a flow channel (a cold water supply pipe **83**) between the hot and cold water mixing part **50** and the cold water supply source **81**.
[0048] The hot and cold water mixing part **50** mixes a hot water that is supplied from the hot water supply source **80** and a cold water that is supplied from the cold water supply source **81**. Specifically, the hot and cold water mixing part **50** switches as to whether or not a hot water is mixed into a cold water. Furthermore, the hot and cold water mixing part **50** controls a rate of a hot water that is mixed into a cold water so as to control a temperature of a mixed water.
[0049] The water mixing faucet unit **20** includes a motor that drives a flow-rate/temperature control part(s). In the present embodiment, the hot and cold water mixing part **50** includes a first motor **51**. The first motor **51** is, for example, a stepping motor. The first motor **51** is driven depending on an operation of a temperature control button **41** of the remote controller **4**, so that the hot and cold water mixing part **50** drives a temperature control valve **210** (see FIG. **7**) so as to switch as to

whether or not a hot water is mixed into a cold water.
[0050] Furthermore, the first motor **51** is driven depending on an operation of the temperature control button **41** of the remote controller **4**, so that the hot and cold water mixing part **50** drives the temperature control valve **210** so as to control a proportion of a hot water that is mixed into a cold water. Furthermore, even in a case where no operation of the temperature control button **41** is executed, for example, as a temperature of a hot water is changed so as to change a temperature of a mixed water, the hot and cold water mixing part **50** controls a ratio of a flow rate of a hot water to a flow rate of a cold water depending on such a temperature of a mixed water, so that it is possible to control such a temperature of a mixed water automatically.
[0051] A hot or cold water flows from the hot and cold water mixing part **50** into the flow rate control part **60**. In a case where a hot or cold water is discharged from a water discharge part(s) **2**, the flow rate control part **60** controls a flow rate of a discharged hot or cold water.
[0052] As described above, the water mixing faucet unit **20** includes a motor that drives a flow-rate/temperature control part(s). In the present embodiment, the flow rate control part **60** includes a second motor **61**. The second motor **61** is, for example, a stepping motor. The second motor **61** is driven depending on a water flow rate control button **42** of the remote controller **4**, so that the flow rate control part **60** drives a flow control valve **230** (see FIG. **7**) so as to control a flow rate of a hot or cold water.
[0053] The water discharge switching part **30** switches discharging or stopping of a hot water or cold water that flows out of the water mixing faucet unit **20**. That is, the water discharge switching part **30** switches discharging or stopping of a hot or cold water from a water discharge part(s) **2**. Furthermore, the water discharge switching part **30** switches a discharging place of a hot or cold water. In the faucet device **1**, switching of discharging or stopping of a hot or cold water from a water discharge part(s) **2** is executed by the water discharge switching part **30** and control of a flow rate of such a hot or cold water in a case where such a hot or cold water is discharged is executed by the flow rate control part **60**.
[0054] The water discharge switching part **30** includes a plurality of electromagnetic valves **31** to **35**. Specifically, the water discharge switching part **30** includes a first electromagnetic valve **31**, a second electromagnetic valve **32**, a third electromagnetic valve **33**, and a fourth electromagnetic valve **34**. The water discharge switching part **30** may include a fifth electromagnetic valve **35**.
[0055] The first electromagnetic valve **31** to the fourth electromagnetic valve **34** are switched to “closed (OFF)” or “open (ON)” depending on an operation of a switching button **43**.
[0056] In a case where the first electromagnetic valve **31** to the fourth electromagnetic valve **34** are “closed”, no hot or cold water is discharged from a water discharge part(s) **2**. That is, in a case where the first electromagnetic valve **31** to the fourth electromagnetic valve **34** are “closed”, a hot or cold water is stopped. In a case where one of the first electromagnetic valve **31** to the fourth electromagnetic valve **34** is “open”, a hot or cold water is discharged from a water discharge part **2** that corresponds to an electromagnetic valve that is “open”.
[0057] The first electromagnetic valve **31** switches discharging or stopping of a hot or cold water in the water tap **10**. The second electromagnetic valve **32** switches discharging or stopping of a hot or cold water in the hand shower **11**. The third electromagnetic valve **33** switches discharging or stopping of a hot or cold water in the overhead shower **12**. The fourth electromagnetic valve **34** switches discharging or stopping of a hot or cold water in the warm pillar **13**.
[0058] For example, in a case where the first electromagnetic valve **31** is “open” and the second electromagnetic valve **32** to the fourth electromagnetic valve **34** are “closed”, a hot or cold water is discharged from the water tap **10**.
[0059] The fifth electromagnetic valve **35** is switched to “closed (OFF)” or “open (ON)” depending on an operation that is executed by an external instrument **89** (for example, a multiple remote controller that is provided in a wash room). Additionally, the fifth electromagnetic valve **35** may be switched to “closed” or “open” depending on an operation of the remote controller **4**. The fifth

electromagnetic valve **35** is a valve for draining a residual water in a hose and/or a pipe of the hand shower **11**, etc., and is normally maintained at "closed". That is, the fifth electromagnetic valve **35** is turned to "closed" only in a case where a residual water process is executed. In a case where the fifth electromagnetic valve **35** is turned to "open", a residual water is drained from a residual water discharge flow channel **88**.

[0060] The controller **7** controls the first motor **51**, the second motor **61**, and the first electromagnetic valve **31** to the fourth electromagnetic valve **34**, depending on an operation of the faucet main body **3** that is received by an operation of the remote controller **4**. Furthermore, the controller **7** controls the fifth electromagnetic valve **35** depending on, for example, an operation of the external instrument **89**.

[0061] The controller **7** includes, for example, a microcomputer that has a Central Processing Unit (CPU), a Read Only Memory (ROM), a Random Access Memory (RAM), etc., and/or various types of circuits. Additionally, the controller **7** may include hardware such as an Application Specific Integrated Circuit (ASIC) and/or a Field Programmable Gate Array (FPGA).

[0062] The communication unit **5** receives an operation signal(s) from the remote controller **4** and the external instrument **89** and transmits a received operation signal(s) to the controller **7**. The remote controller **4** and the external instrument **89** are connected to the communication unit **5** by wired communication or wireless communication. The controller **7** is connected to the communication unit **5** by wired communication or wireless communication.

[0063] Next, a faucet main body **3** will be explained with reference to FIG. **4**. FIG. **4** is a perspective view that illustrates a faucet main body **3**. The faucet main body **3** includes a flow channel unit **70** in addition to a water mixing faucet unit **20**, a water discharge switching part **30**, and a controller **7** (see FIG. **3**). The faucet main body **3** is attached to a wall part **1004***a* of a bathroom unit **1000** (see FIG. **1**) by the flow channel unit **70**.

[0064] A flow channel where a hot water that is supplied from a hot water supply source **80** (see FIG. **3**) flows into the water mixing faucet unit **20**, a flow channel where a cold water that is supplied from a cold water supply source **81** (see FIG. **3**) flows into the water mixing faucet unit **20**, and a flow channel where a hot or cold water flows from the water mixing faucet unit **20** into each water discharge part **2** are formed in the flow channel unit **70**. Additionally, the water discharge switching part **30** is provided in the flow channel unit **70**. Furthermore, water shut-off valves **84**, **85** (see FIG. **3**) are provided in the flow channel unit **70**. Furthermore, a residual water discharge flow channel **88** is provided in the flow channel unit **70**.

[0065] In the faucet main body **3**, the flow channel unit **70** is provided at a back side thereof and the water mixing faucet unit **20** is provided at a front side of the flow channel unit **70**. Additionally, a part of the flow channel unit **70** is provided at an upper side of the water mixing faucet unit **20**. Furthermore, a controller case **201** that is a second waterproof case as described later and houses the controller **7** at a front side of the water mixing faucet unit **20** is provided therein.

[0066] Next, a water mixing faucet unit **20** will be explained with reference to FIG. **5** to FIG. **7**. FIG. **5** is a perspective view that illustrates a water mixing faucet unit **20**. FIG. **6** is a front view that illustrates a water mixing faucet unit **20**. FIG. **7** is a perspective view in a cross section along VII-VII in FIG. **6**.

[0067] The water mixing faucet unit **20** is provided so as to extend in leftward and rightward directions. A first motor case **200***a* that is a first waterproof case as described later and houses a first motor **51** of a hot and cold water mixing part **50** is provided at one end of the water mixing faucet unit **20** in leftward and rightward directions, specifically, a left end thereof. A second motor case **200***b* that is a first waterproof case as described later and houses a second motor **61** of a flow rate control part **60** is provided at another end of the water mixing faucet unit **20** in leftward and rightward directions, specifically, a right end thereof.

[0068] The hot and cold water mixing part **50** moves a temperature control valve **210** (a control valve) in leftward and rightward directions so as to switch as to whether or not a hot water that is supplied from a hot water supply channel **203** is mixed into a cold water that is supplied from a cold water supply channel **202**.

[0069] For example, the temperature control valve **210** contacts a case **211** that is provided at a left side of the temperature control valve **210**, so that supply of a hot water is shut off by the temperature control valve **210** and thereby such a hot water is not mixed into a cold water that is supplied from the cold water supply channel **202**.

[0070] Furthermore, the temperature control valve **210** is separated from the case **211** and thereby supply of a hot water is not shut off by the temperature control valve **210**, so that a hot water that is supplied from the hot water supply channel **203** is mixed into a cold water that is supplied from the cold water supply channel **202**.

[0071] The temperature control valve **210** is pressed by a temperature-sensitive spring **212** in a direction where it is separated from the case **211**. Furthermore, the temperature control valve **210** is pressed by a bias spring **213** in a direction where it contacts the case **211**. That is, the temperature control valve **210** is moved in leftward and rightward directions according to a magnitude relationship between a pressing force of the temperature-sensitive spring **212** and a pressing force of the bias spring **213**.

[0072] The temperature-sensitive spring **212** is a spring with a spring constant that is changed depending on a temperature thereof and is composed of, for example, a shape memory alloy. The bias spring **213** is a spring with a spring constant that is nearly constant with respect to a temperature thereof.

[0073] In a case where a pressing force of the bias spring **213** is greater than a pressing force of the temperature-sensitive spring **212**, the temperature control valve **210** is moved toward a left side that is a side of the case **211**. Additionally, in a case where the temperature control valve **210** contacts the case **211**, the temperature control valve **210** is held in a state where it contacts the case **211**.

[0074] In a case where a pressing force of the temperature-sensitive spring **212** is greater than a pressing force of the bias spring **213**, the temperature control valve **210** is moved toward a right side in such a manner that it is separated from the case **211**.

[0075] In a case where a pressing force of the temperature-sensitive spring **212** and a pressing force of the bias spring **213** are equal, the temperature control valve **210** is not moved in a leftward or rightward direction and is held at a position where such a pressing force of the temperature-sensitive spring **212** and such a pressing force of the bias spring **213** are balanced.

[0076] The temperature-sensitive spring **212** contacts a liner **214** at another end that is an opposite side of one end that contacts the temperature control valve **210**. The liner **214** is connected to a rotational shaft part **216** through a spindle **215**. The spindle **215** is supported by the case **211** so as to be rotatable and converts a rotational movement of the rotational shaft part **216** to a translatory movement of the liner **214** in leftward and rightward directions. Hence, the liner **214** is moved in leftward and rightward directions depending on rotation of the rotational shaft part **216**.

[0077] The rotational shaft part **216** is connected to the first motor **51** through a transmission mechanism. A transmission mechanism is housed in the first motor case **200***a*. A transmission mechanism includes a first gear that is connected to a rotational shaft of the first motor **51** and a second gear that is connected to the rotational shaft part **216**. A first gear and A second gear are engaged with one another. Rotation of a rotational shaft of the first motor **51** is transmitted to the rotational shaft part **216** through a transmission mechanism. Additionally, such a transmission mechanism is similar to a transmission mechanism in the flow rate control part **60** as described later.

[0078] In the hot and cold water mixing part **50**, the liner **214** is moved in leftward and rightward directions according to a rotational position of the rotational shaft part **216**, that is, a rotational position of a rotational shaft of the first motor **51**. That is, it is possible for the hot and cold water mixing part **50** to change a position of the temperature control valve **210** at a position where a pressing force of the temperature-sensitive spring **212** and a pressing force of the bias spring **213** are balanced, according to a rotational position of a rotational shaft of the first motor **51**. Thereby, it is possible for the hot and cold water mixing part **50** to cause a temperature of a mixed water to be a set temperature that corresponds to a rotational position of a rotational shaft of the first motor **51**, in a case where such a mixed water is discharged.

[0079] Furthermore, in a case where a mixed water is discharged, for example, as a temperature of a hot water is changed so as to change a temperature of such a mixed water, the temperature-sensitive spring **212** is expanded or compressed depending on a temperature of such a mixed water and thereby the temperature control valve **210** is moved in leftward and rightward directions, so that a balanced position of the temperature control valve **210** is changed automatically. Thereby, an amount of a hot water and an amount of a cold water in a mixed water is controlled, so that a temperature of such a mixed water is controlled automatically.

[0080] The hot and cold water mixing part **50** is communicated with the flow rate control part **60** through a communication hole **220**. A hot or cold water flows from the hot and cold water mixing part **50** into the flow rate control part **60** through the communication hole **220**.

[0081] The flow rate control part **60** rotates a flow control valve **230** (a control valve) around a virtual axis that extends in leftward and rightward directions, as a center thereof, so as to change an area where an opening of the flow control valve **230** is communicated with an opening of a control member **221** and thereby change an opening degree of the flow control valve **230**. The flow rate control part **60** changes an opening degree of the flow control valve **230** so as to control a flow rate of a hot or cold water that flows out of the water mixing faucet unit **20**, that is, a flow rate of a hot or cold water that is discharged from a water discharge part(s) **2** (see FIG. **3**).

[0082] Next, a faucet main body **3** will further be explained with reference to FIGS. **8** to **13**. FIG. **8** is a perspective view of a faucet main body **3**. FIG. **9** is a perspective view that illustrates first waterproof cases **200***a*, **200***b* and a second waterproof case **201**. FIG. **10** is a perspective view that illustrates a variation of first waterproof cases **200***a*, **200***b* and a second waterproof case **201**. FIG. **11** is a cross-sectional view that illustrates a binding member **306** and a sealing member **307**. FIG. **12** is a perspective view that illustrates a binding member **306** and a sealing member **307**. FIG. **13** is a perspective view that illustrates a cover member **400**.

[0083] As described above, the faucet main body **3** is arranged inside a bathroom unit **1000** (see FIG. **1**). As illustrated in FIG. **8**, in the faucet main body **3**, a first motor **51** that drives a hot and cold water mixing part **50** is housed in a first motor case **200***a* that is a first waterproof case. In the faucet main body **3**, a second motor **61** that drives a flow rate control part **60** is housed in a second motor case **200***b* that is a first waterproof case. That is, a first waterproof case(s) include(s) the first motor case **200***a* and the second motor case **200***b*.

[0084] As illustrated in FIG. **9**, the first motor case **200***a* houses the first motor **51** so as to isolate the first motor **51** from an internal space of the bathroom unit **1000**. The second motor case **200***b* houses the second motor **61** so as to isolate the second motor **61** from an internal space of the bathroom unit **1000**. Thus, the first motor case **200***a* and the second motor case **200***b* that are first waterproof cases house the first motor **51** and the second motor **61** separately.

[0085] Furthermore, in the faucet main body **3**, a controller **7** is housed in the controller case **201** that is a second waterproof case. The controller case **201** houses the controller **7** so as to isolate the controller **7** from an internal space of the bathroom unit **1000**.

[0086] One waterproof case among the first motor case **200***a* and the second motor case **200***b* that are first waterproof cases and the controller case **201** that is a second waterproof case is communicated with an external space of the bathroom unit **1000**.

[0087] Furthermore, another waterproof case among the first motor case **200***a* and the second motor case **200***b* that are first waterproof cases and the controller case **201** that is a second waterproof case is communicated with one waterproof case that is communicated with an external space of the bathroom unit **1000**.

[0088] In an example as illustrated in FIG. **9**, the controller case **201** that is a second waterproof

case is communicated with an external space of the bathroom unit **1000** and the first motor case **200***a* and the second motor case **200***b* that are first waterproof cases are communicated with the controller case **201**. The first motor **51** is arranged closer to a part that is communicated with an external space of the bathroom unit **1000** than the second motor **61**.
[0089] By returning to FIG. **8**, one waterproof case among the first motor case **200***a* and the second motor case **200***b* that are first waterproof cases and the controller case **201** that is a second waterproof case is communicated with an external space of the bathroom unit **1000**. In such a case, one waterproof case is communicated with an external space of the bathroom unit **1000** through a hose member **300**.
[0090] The hose member **300** has flexibility. One end part of the hose member **300** is connected to the faucet main body **3** and another end part thereof is connected to a wall (a wall surface WS) of the bathroom unit **1000**. One waterproof case among the first motor case **200***a* and the second motor case **200***b* that are first waterproof cases and the controller case **201** that is a second waterproof case (for example, the controller case **201** that is a second waterproof case) is communicated with a space behind a wall of the bathroom unit **1000**.
[0091] As illustrated in FIG. **9**, in a case where an air inside a waterproof case flows out to an external space of the bathroom unit **1000**, an air Al from the first motor case **200***a* as a first waterproof case flows through the hose member **300** from a communication hole **301** toward a wall surface WS through the controller case **201** that is a second waterproof case. Additionally, the communication hole **301** is a hole for communicating between the first motor case **200***a* as a first waterproof case and the controller case **201** that is a second waterproof case.
[0092] Furthermore, in a case where an air inside a waterproof case flows out to an external space of the bathroom unit **1000**, an air A**2** from the second motor case **200***b* as a first waterproof case flows out from a communication hole **302** to the hose member **300** toward a wall surface WS through the controller case **201** that is a second waterproof case. Additionally, the communication hole **302** is a hole for communicating between the second motor case **200***b* as a first waterproof case and the controller case **201** that is a second waterproof case.
[0093] An air A**3** that is provided by combining an air A**1** from the first motor case **200***a* and an air A**2** from the second motor case **200***b* flows out from the hose member **300** to an external space of the bathroom unit **1000**.
[0094] Furthermore, in a case where an air flows from an external space of the bathroom unit **1000** into an inside of a waterproof case, an air A**4** from such an external space of the bathroom unit **1000** flows into the hose member **300**, and flows out from the hose member **300** to the first motor case **200***a* and the second motor case **200***b* that are first waterproof cases through the controller case **201** that is a second waterproof case.
[0095] In such a configuration, the first waterproof cases **200***a*, **200***b* and the second waterproof case **201** are communicated with an external space of the bathroom unit **1000** such as a space behind a wall thereof, so that an air A**4** outside the bathroom unit **1000** flows into the first waterproof case **200***a* and the second waterproof case **201** and an air A**3** flows out from the first waterproof cases **200***a*, **200***b* and the second waterproof case **201** to an outside of the bathroom unit **1000**. An air A**4** outside the bathroom unit **1000** moves in and out of the first waterproof cases **200***a*, **200***b* and the second waterproof case **201**, so that it is possible to cause the first waterproof cases **200***a*, **200***b* and the second waterproof case **201** to execute so-called "breathing". Thus, the first waterproof cases **200***a*, **200***b* and the second waterproof case **201** execute "breathing", so that it is possible to cause a temperature inside the first waterproof cases **200***a*, **200***b* and the second waterproof case **201** to be identical to be a temperature outside the bathroom unit **1000**. Thereby, it is possible to reduce or prevent occurrence of dew condensation inside the first waterproof cases **200***a*, **200***b* and the second waterproof case **201**. Furthermore, the motors **51**, **61** are covered by the first waterproof cases **200***a*, **200***b* and the controller **7** is covered by the second waterproof case **201**, that is, the motors **51**, **61** and the controller **7** are covered by waterproof cases separately, so

that it is possible to minimize sizes of respective waterproof cases. Thereby, it is possible to reduce or prevent a size increase of the faucet main body **3**.

[0096] Furthermore, one of the first waterproof cases **200***a*, **200***b* and the second waterproof case **201** is communicated with an external space of the bathroom unit **1000** and another thereof is communicated with the one, so that another waterproof case is indirectly communicated with such an external space of the bathroom unit **1000** through one waterproof case. Thereby, it is possible to reduce a number of a hole(s) that is/are formed on a wall of the bathroom unit **1000**, etc., in order to communicate a waterproof case with an external space of the bathroom unit **1000**, so that it is possible to increase a degree of freedom of arrangement of the motors **51**, **61** and the controller **7**. Thereby, it is possible to reduce or prevent a size increase of the faucet main body **3**. Furthermore, in a case where areas of holes that are formed on a wall of the bathroom unit **1000**, etc., are identical, an efficiency of air ventilation is increased for a single large hole rather than for a plurality of small holes, so that it is possible to reduce or prevent occurrence of dew condensation effectively. Furthermore, a single large hole is provided, so that only single connection is needed as compared with a case where a plurality of small holes are connected separately, and thereby, manufacturability and workability thereof are improved.

[0097] Furthermore, the first waterproof cases **200***a*, **200***b* house the first motor **51** and the second motor **61** separately, the second waterproof case **201** is communicated with an external space of the bathroom unit **1000**, and further the first motor **51** is arranged closer to a part that is communicated with such an external space of the bathroom unit **1000** than the second motor **61**, so that even in a case where the first waterproof cases **200***a*, **200***b* that house the motors **51**, **61** are indirectly communicated with such an external space of the bathroom unit **1000**, it is possible to arrange the first motor **51** for temperature control for a hot or cold water that causes dew condensation more readily than the second motor **61** for flow rate control, closer to a part that is communicated with such an external space of the bathroom unit **1000** such as a wall of the bathroom unit **1000**. Thereby, it is possible to reduce or prevent occurrence of dew condensation effectively. Furthermore, replacement of the first motor **51** is also facilitated.

[0098] Furthermore, a waterproof case that is communicated with an external space of the bathroom unit **1000** among the first waterproof cases **200***a*, **200***b* and the second waterproof case **201** is communicated with such an external space of the bathroom unit **1000** through the hose member **300**, so that a degree of freedom of arrangement of the motors **51**, **61** and the controller **7** is increased. In particular, the hose member **300** has flexibility that allows to be bent freely to certain extent, so that a degree of freedom of arrangement of a waterproof case(s) is increased. Furthermore, the hose member **300** has flexibility, so that it is possible to ensure watertightness without applying a force to a waterproof case(s).

[0099] Furthermore, by returning to FIG. **8**, the second motor case **200***b* that houses the second motor **61** is connected to the controller case **201** that is a second waterproof case and houses the controller **7** by a hose member **310**. The second motor case **200***b* is communicated with an external space of the bathroom unit **1000**, that is, a space behind a wall of the bathroom unit **1000**, through the hose member **300**, the controller case **201**, and the hose member **310**.

[0100] The hose member **310** has flexibility like the hose member **300** as described above. Thereby, a degree of freedom of arrangement of the second motor **61** and the controller **7** is increased. Also in such a case, the hose member **310** has flexibility that allows to be bent freely to certain extent, like the hose member **300** as described above, so that a degree of freedom of arrangement of a waterproof case(s) is increased. Furthermore, it is possible to ensure watertightness without applying a force to a waterproof case(s).

[0101] Furthermore, a harness **303** (see FIG. **11**) is inserted through the communication holes **301**, **302**, so as to pass through these communication holes **301**, **302**. The harness **303** connects a motor(s) (the first motor **51** and the second motor **61**) and the controller **7** electrically.

[0102] In such a configuration, in a case where the harness **303** is used in order to connect the motors **51**, **61** and the controller **7** electrically, no dedicated hole for passing the harness **303** therethrough is needed. Thereby, it is possible to complete electrical connection of the motors **51**, **61** and the controller **7** inside the first waterproof cases **200***a*, **200***b* and the second waterproof case **201**, so that it is possible to improve a waterproof property thereof.

[0103] Furthermore, as illustrated in FIG. **10**, in a case where the first motor case **200***a* and the second motor case **200***b* that are first waterproof cases house the first motor **51** and the second motor **61** separately, for example, the first motor case **200***a* that houses the first motor **51** among such first waterproof cases may be communicated with an external space of the bathroom unit **1000** (see FIG. **1**).

[0104] In such a configuration, it is possible to arrange the first motor **51** for temperature control for a mixed hot and cold water where dew condensation is readily caused near a cold water at a low temperature because a hot water at a high temperature and such a cold water at a low temperature flow thereinto more than the second motor **61** for flow rate control, closer to a part that is communicated with an external space of the bathroom unit **1000** such as a wall of the bathroom unit **1000**. Thereby, it is possible to reduce or prevent occurrence of dew condensation effectively.

[0105] As illustrated in FIG. **11** and FIG. **12**, the controller case **201** that is a second waterproof case has a harness hole **305** where a plurality of harnesses **304** are inserted therethrough. The plurality of harnesses **304** electrically connect each of a plurality of electromagnetic valves that switch flow channels where a hot or cold water flows therethrough and the controller **7**. The plurality of harnesses **304** are configured to be bundled by a binding member **306**. The binding member **306** is, for example, a cabtyre that covers the plurality of harnesses **304** with a rubber, etc.

[0106] Furthermore, the harness hole **35** where the plurality of harnesses **304** are inserted therethrough includes a sealing member **307**. In a state where the harnesses **304** are inserted through the harness hole **35**, a gap between the harness hole **35** and the harnesses **304** are sealed by the sealing member **307** so as to be watertight. The sealing member **307** is, for example, a grommet.

[0107] In such a configuration, the plurality of harnesses **304** are bundled by a binding member (a cabtyre) **306** and inserted through the harness hole **305**, so that it is possible to reduce a route where a water flows to the controller **7**. Furthermore, although a large harness hole **305** is needed in a case where the plurality of harnesses **304** are bundled, the harness hole **305** is closed by the sealing member (a grommet) **307**, so that it is possible to reduce or prevent intruding of a water into an inside(s) of a waterproof case(s) through a harness(es) **304** even in such a case.

[0108] As illustrated in FIG. **13**, the faucet main body **3** further includes a cover member **400**. The cover member **400** is a soft member. The cover member **400** covers a first waterproof case(s) (the first motor case **200***a* and the second motor case **200***b*) and a second waterproof case (the controller case **201**), for example from a front side thereof.

[0109] In such a configuration, it is possible to reduce or prevent shifting of a relative position of the first waterproof cases **200***a*, **200***b* and the second waterproof case **201** that is caused by, for example, a worker that contacts a waterproof case(s), at a time of construction and a time of maintenance, etc., and is not allowed to ensure a waterproof property thereof. Herein, the cover member **400** is soft, so that it is possible to reduce or prevent gripping of the cover member **400** by a worker more effectively.

[0110] Furthermore, the cover member **400** may be provided with, for example, a display label **401** for a worker for construction. It is preferable to provide the display label **401** at a prominent position such as a front surface and/or an upper surface of the cover member **400**. For example, a message and/or an illustration for attracting attention of a worker so as to prevent shifting of a position(s) of a waterproof case(s) is/are displayed on the display label **401**.

[0111] An aspect of an embodiment aims to provide a faucet device that is capable of reducing or preventing occurrence of dew condensation inside a waterproof case that houses a motor and a controller and is capable of reducing or preventing a size increase of a faucet main body.

[0112] A faucet device according to an aspect of an embodiment is a faucet device including a water discharge part that discharges a hot or cold water, and a faucet main body that is arranged inside a bathroom and supplies a hot or cold water to the water discharge part, wherein the faucet main body includes a flow-rate/temperature control part that executes flow rate control and temperature control for a hot or cold water that is discharged from the water discharge part, a motor that drives the flow-rate/temperature control part, a controller that controls driving of the motor, a first waterproof case that houses the motor and isolates the motor from an internal space of the bathroom, and a second waterproof case that houses the controller and isolates the controller from an internal space of the bathroom, and the first waterproof case and the second waterproof case are communicated with an external space of the bathroom.
[0113] In such a configuration, a first waterproof case and a second waterproof case are communicated with an external space of a bathroom (for example, a space behind a wall of a bathroom), so that an air outside such a bathroom flows into such a first waterproof case and a second waterproof case and an air flows from such a first waterproof case and a second waterproof case to an outside of such a bathroom. An air outside a bathroom flows into or from a first waterproof case and a second waterproof case, so that it is possible to cause such a first waterproof case and a second waterproof case to execute so-called “breathing”. Thus, a first waterproof case and a second waterproof case execute “breathing”, so that it is possible to cause a temperature inside such a first waterproof case and a second waterproof case to be identical to a temperature outside a bathroom. Thereby, it is possible to reduce or prevent occurrence of dew condensation inside a first waterproof case and a second waterproof case. Furthermore, a motor is covered by a first waterproof case and a controller is covered by a second waterproof case, that is, such a motor and such a controller are covered by such waterproof cases separately, so that it is possible to minimize a size of each waterproof case. Thereby, it is possible to reduce or prevent a size increase of a faucet main body.
[0114] Furthermore, in the faucet device as described above, the first waterproof case and the second waterproof case are provided in such a manner that one of the waterproof cases is communicated with an external space of the bathroom and another of the waterproof cases is communicated with the one of the waterproof cases.
[0115] In such a configuration, one of a first waterproof case and a second waterproof case is communicated with an external space of a bathroom and another thereof is communicated with the one, so that such another of waterproof cases is indirectly communicated with such an external space of a bathroom through such one of waterproof cases. Thereby, it is possible to reduce a number of a hole(s) that is/are formed on a wall of a bathroom, etc., in such a manner that a waterproof case is communicated with an external space of such a bathroom, so that it is possible to increase a degree of freedom of arrangement of a motor and a controller. Thereby, it is possible to reduce or prevent a size increase of a faucet main body. Furthermore, in a case where cross sections of holes that are formed on a wall of a bathroom, etc., are identical, an efficiency of air ventilation is increased for a single large hole rather than for a plurality of small holes, so that it is possible to reduce or prevent occurrence of dew condensation effectively.
[0116] Furthermore, the faucet device as described above includes a harness that connects the motor and the controller, wherein the harness passes through a communication hole for communicating between the first waterproof case and the second waterproof case.
[0117] In such a configuration, in a case where a harness is used in order to connect a motor and a controller electrically, no dedicated hole for passing such a harness therethrough is needed. Thereby, it is possible to complete electrical connection of a motor and a controller inside a first waterproof case and a second waterproof case, so that a waterproof property thereof is improved.
[0118] Furthermore, in the faucet device as described above, the motor has a first motor for temperature control for a hot or cold water and a second motor for flow rate control for a hot or cold water, the first waterproof case houses the first motor and the second water separately, the

second waterproof case is communicated with an external space of the bathroom, and the first motor is arranged closer to a part that is communicated with an external space of the bathroom than the second motor.

[0119] In such a configuration, even in a case where a first waterproof case that houses a motor, among such a first waterproof case and a second waterproof case, is indirectly communicated with an external space of a bathroom, it is possible to arrange a first motor for temperature control for a hot or cold water that causes dew condensation more readily than a second motor for flow rate control, close to a part that is communicated with such an external space of a bathroom, such as a wall of such a bathroom. Thereby, it is possible to reduce or prevent occurrence of dew condensation effectively. Furthermore, replacement of a first motor is also facilitated.

[0120] Furthermore, in the faucet device as described above, the motor has a first motor for temperature control for a hot or cold water and a second motor for flow rate control for a hot or cold water, the first waterproof case includes a first motor case that houses the first motor and a second motor case that houses the second motor, and the first motor case is communicated with an external space of the bathroom.

[0121] In such a configuration, it is possible to arrange a first motor for temperature control for a hot or cold water that causes dew condensation more readily than a second motor for flow rate control, close to a part that is communicated with an external space of a bathroom, such as a wall of such a bathroom. Thereby, it is possible to reduce or prevent occurrence of dew condensation effectively.

[0122] Furthermore, in the faucet device as described above, the first waterproof case and the second waterproof case are provided in such a manner that one of the waterproof cases is communicated with an external space of the bathroom and another of the waterproof cases is communicated with the one of the waterproof cases, and the one of the waterproof cases is communicated with an external space of the bathroom through a hose member.

[0123] In such a configuration, a waterproof case that is communicated with an external space of a bathroom, among a first waterproof case and a second waterproof case, is communicated with such an external space of a bathroom through a hose member, so that a degree of freedom of arrangement of a motor and a controller is increased. In particular, a hose member has flexibility that allows to be bent freely to certain extent, so that it is possible to increase a degree of freedom of arrangement of a waterproof case(s). Furthermore, a hose member has flexibility, so that it is possible to ensure watertightness without applying a force to a waterproof case(s).

[0124] Furthermore, the faucet device as described above includes a plurality of electromagnetic valves that switch between flow channels where a hot or cold water flows therethrough, and a plurality of harnesses that connect each of the plurality of electromagnetic valves and the controller, wherein the second waterproof case has a harness hole where the harnesses are inserted therethrough, the plurality of harnesses are configured to be bundled by a binding member and to be inserted through the harness hole, and the harness hole includes a sealing member that seals a gap between the harness hole and the harnesses to be watertight in a state where the harnesses are inserted therethrough.

[0125] In such a configuration, a plurality of harnesses are bundled by a binding member (a cabtyre) and are inserted through a harness hole, so that it is possible to reduce a route where a water flows to a controller. Furthermore, although a large harness hole is needed in a case where a plurality of harnesses are bundled, a harness hole is closed by a sealing member (a grommet) even in such a case, so that it is possible to reduce or prevent intruding of a water into an inside(s) of a waterproof case(s) through such a harness(es).

[0126] Furthermore, the faucet device as described above includes a cover member that covers the first waterproof case and the second waterproof case.

[0127] In such a configuration, it is possible to reduce or prevent shifting of a relative position of a first waterproof case and a second waterproof case that is caused by, for example, a worker that contacts a waterproof case(s), at a time of construction and a time of maintenance, etc., and is not allowed to ensure a waterproof property thereof. Herein, a cover member is soft, so that it is possible to reduce or prevent gripping of a cover member by a worker more effectively.

[0128] For a faucet device according to an aspect of an embodiment, it is possible to reduce or prevent occurrence of dew condensation inside a waterproof case that houses a motor and a controller and to reduce or prevent a size increase of a faucet main body.

APPENDIX

[0129] (1) A faucet device, including: [0130] a water discharge part that discharges a hot or cold water; and [0131] a faucet main body that is arranged inside a bathroom and supplies a hot or cold water to the water discharge part, wherein [0132] the faucet main body includes: [0133] a flow-rate/temperature control part that executes flow rate control and temperature control for a hot or cold water that is discharged from the water discharge part; [0134] a motor that drives the flow-rate/temperature control part; [0135] a controller that controls driving of the motor; [0136] a first waterproof case that houses the motor and isolates the motor from an internal space of the bathroom; and [0137] a second waterproof case that houses the controller and isolates the controller from an internal space of the bathroom, and [0138] the first waterproof case and the second waterproof case are communicated with an external space of the bathroom. [0139] (2) The faucet device according to (1), wherein [0140] the first waterproof case and the second waterproof case are provided in such a manner that one of the waterproof cases is communicated with an external space of the bathroom and another of the waterproof cases is communicated with the one of the waterproof cases. [0141] (3) The faucet device according to (2), including [0142] a harness that connects the motor and the controller, wherein [0143] the harness passes through a communication hole for communicating between the first waterproof case and the second waterproof case. [0144] (4) The faucet device according to (2) or (3), wherein [0145] the motor has a first motor for temperature control for a hot or cold water and a second motor for flow rate control for a hot or cold water, [0146] the first waterproof case houses the first motor and the second water separately, [0147] the second waterproof case is communicated with an external space of the bathroom, and [0148] the first motor is arranged closer to a part that is communicated with an external space of the bathroom than the second motor. [0149] (5) The faucet device according to (2) or (3), wherein [0150] the motor has a first motor for temperature control for a hot or cold water and a second motor for flow rate control for a hot or cold water, [0151] the first waterproof case includes a first motor case that houses the first motor and a second motor case that houses the second motor, and [0152] the first motor case is communicated with an external space of the bathroom. [0153] (6) The faucet device according to any one of (1) to (5), wherein [0154] the first waterproof case and the second waterproof case are provided in such a manner that one of the waterproof cases is communicated with an external space of the bathroom and another of the waterproof cases is communicated with the one of the waterproof cases, and [0155] the one of the waterproof cases is communicated with an external space of the bathroom through a hose member. [0156] (7) The faucet device according to any one of (1) to (6), including: [0157] a plurality of electromagnetic valves that switch between flow channels where a hot or cold water flows therethrough; and [0158] a plurality of harnesses that connect each of the plurality of electromagnetic valves and the controller, wherein [0159] the second waterproof case has a harness hole where the harnesses are inserted therethrough, [0160] the plurality of harnesses are configured to be bundled by a binding member and to be inserted through the harness hole, and [0161] the harness hole includes a sealing member that seals a gap between the harness hole and the harnesses to be watertight in a state where the harnesses are inserted therethrough. [0162] (8) The faucet device according to any one of (1) to (7), including [0163] a cover member that covers the first waterproof case and the second waterproof case.

[0164] It is possible for a person(s) skilled in the art to readily derive an additional effect(s) and/or variation(s). Hence, a broader aspect(s) of the present invention is/are not limited to a specific detail(s) and a representative embodiment(s) as illustrated and described above. Therefore, various modifications are possible without departing from the spirit or scope of a general inventive concept that is defined by the appended claim(s) and an equivalent(s) thereof.

Claims

1. A faucet device, comprising: a water discharge part that discharges a hot or cold water; and a faucet main body that is arranged inside a bathroom and supplies a hot or cold water to the water discharge part, wherein the faucet main body includes: a flow-rate/temperature control part that executes flow rate control and temperature control for a hot or cold water that is discharged from the water discharge part; a motor that drives the flow-rate/temperature control part; a controller that controls driving of the motor; a first waterproof case that houses the motor and isolates the motor from an internal space of the bathroom; and a second waterproof case that houses the controller and isolates the controller from an internal space of the bathroom, and the first waterproof case and the second waterproof case are communicated with an external space of the bathroom.
2. The faucet device according to claim 1, wherein the first waterproof case and the second waterproof case are provided in such a manner that one of the waterproof cases is communicated with an external space of the bathroom and another of the waterproof cases is communicated with the one of the waterproof cases.
3. The faucet device according to claim 2, comprising a harness that connects the motor and the controller, wherein the harness passes through a communication hole that communicates between the first waterproof case and the second waterproof case.
4. The faucet device according to claim 2, wherein the motor has a first motor for temperature control for a hot or cold water and a second motor for flow rate control for a hot or cold water, the first waterproof case houses the first motor and the second water separately, the second waterproof case is communicated with an external space of the bathroom, and the first motor is arranged closer to a part that is communicated with an external space of the bathroom than the second motor.
5. The faucet device according to claim 2, wherein the motor has a first motor for temperature control for a hot or cold water and a second motor for flow rate control for a hot or cold water, the first waterproof case includes a first motor case that houses the first motor and a second motor case that houses the second motor, and the first motor case is communicated with an external space of the bathroom.
6. The faucet device according to claim 1, wherein the first waterproof case and the second waterproof case are provided in such a manner that one of the waterproof cases is communicated with an external space of the bathroom and another of the waterproof cases is communicated with the one of the waterproof cases, and the one of the waterproof cases is communicated with an external space of the bathroom through a hose member.
7. The faucet device according to claim 1, comprising: a plurality of electromagnetic valves that switch between flow channels where a hot or cold water flows therethrough; and a plurality of harnesses that connect each of the plurality of electromagnetic valves and the controller, wherein the second waterproof case has a harness hole where the harness is inserted therethrough, the plurality of harnesses are configured to be bundled by a binding member and to be inserted through the harness hole, and the harness hole includes a sealing member that seals a gap between the harness hole and the harness to be watertight in a state where the harness is inserted therethrough.
8. The faucet device according to claim 1, comprising a cover member that covers the first waterproof case and the second waterproof case.